

The Self-Referential Log-Encoded Polar LUT Paradigm

mosTADPOLE (the green one)

OTAN2 / f8 / Det-0 / WElip

A packed, relational execution fabric

Data, operators and digital footprints share a common word structure. A self-referential log-encoded polar lookup system drives relative phase traversal, generative topology, mirrored execution and bounded active-state eviction.

Architecture, operator contracts, state transitions,
reference packing and reproducible execution checks.

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How the specification is organized

The source develops the architecture in successive stages. This edition preserves its named primitives and makes the source’s final emphasis—the self-referential log-encoded polar LUT—the organizing center. Source references identify the relevant pages rather than treating the successive conversational descriptions as one already fixed implementation.

Definitions and formal conventions introduce the notation, contracts and explicit design choices made in this edition. **Requirements** state what a realization must supply or preserve. **Properties** are derived under their stated conditions. RP32 is the concrete arithmetic reference profile introduced here; it does not stand in for unspecified physical measurements or growth rules.

1	Paradigm charter	3	10	FIFO eviction and regeneration	12
2	State and execution model	4	11	Mirrored agents and time	13
3	Log-polar core and OTAN2	5	12	LUS, DIGID and UU ID	14
4	Klein topology and Lie structure	6	13	WElip forward-time profile	15
5	Fields, axes and cartography	7	14	RGBA realization and scaling	16
6	Same-structure words	8	15	Wave and sensory adapters	17
7	RP32 reference encoding	9	16	Transition contract	18
8	L-system organogram	10	17	Derived properties	19
9	f8 indexing and Hadamard	11	18	Conformance and checks	20
Appendix A Worked packed-state trace					21
Appendix B Reference arithmetic					22
Appendix C Source concordance					23
Appendix D Realization manifest and attribution					24

1 Paradigm charter

Source basis: [S], pp. 4–5, 8–14, 25–26 and 32–37.

Central thesis. The computational object is a self-referential, log-encoded polar packed state. Spatial nodes, operators, packets and log footprints are interpretations of the same carrier; not separate foundational architectures.

The author's source correction places the entire architecture around a LUT that is “self referential and packed” [S, p. 25]. The later deployment descriptions are therefore treated as realizations of that core, rather than replacements by ordinary image-grid semantics.

Architectural commitments

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- C1 Relational phase.** OTAN2 operates on relative radial-sweep phase, without a mandatory global physical origin or fixed Cartesian axes.
 - C2 Uniform carrier.** Nodes, instructions, environmental samples and digital footprints use profile-defined packed words.
 - C3 Self-reference.** A LUT result can determine the interpretation and next lookup of subsequent packed states.
 - C4 Topological traversal.** Klein-bottle identifications, the Ψ traversal relation and f8 indexing organize admissible movement.
 - C5 Generative structure.** An L-system organogram derives states from baseline data, production rules, time and admitted inputs.
 - C6 Eviction as an operation.** A bounded FIFO drop table releases active derived states; regeneration follows a declared derivation.
 - C7 Double-packed execution.** Paired representations maintain an explicit mirror relation through updates and seam transitions.
 - C8 Forward progression.** LUS emission, DIGID footprints and the WElip profile are temporal interpretations of the same execution fabric.
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Meaning of the paradigm's scope

Definition 1. *Coordinate-free semantics* means that the physical interpretation depends on relative relations and topological equivalence classes, not on the arbitrary labels used to encode them. Changing a local representation must preserve the represented transition.

Formal convention 1. The source's “universal” direction is expressed here as *adapter-independent packet semantics*. Its “infinite” direction is expressed as operation at any finite logical tick with a declared active-state budget. Its f8 “BVH replacement / S2 superseder” language identifies the intended indexing role [S, pp. 4–5], with comparative performance a separate measurable design target.

2 State and execution model

Source basis: [S], pp. 8–12, 16–18 and 33–37.

Definition 2. A versioned realization is a tuple

$$\mathcal{A}_v = (\mathcal{W}_v, \mathcal{L}_v, \mathcal{P}_v, \mathcal{H}_v, \mathcal{I}_v, \Pi_v),$$

where $\mathcal{W}_v$ is the packed-word carrier, $\mathcal{L}_v$ the log-polar lookup family, $\mathcal{P}_v$ the production grammar, $\mathcal{H}_v$ the topological interpretation, $\mathcal{I}_v$ the interpreter, and Π_v the realization manifest.

A runtime state is represented by

$$S_T = (v, \omega, T, X_T, Q_T, C_T, \mathcal{E}_{\leq T}, \mathcal{D}_T).$$

ω	Baseline / genesis-axiom data, with a declared version and retained contents.
T	Logical tick, including an epoch when a finite clock field wraps.
X_T	Current packed operators, twin-agent states and active interpretation context.
Q_T, C_T	Ordered active FIFO and its capacity, measured in the manifest's word or pair units.
$\mathcal{E}_{\leq T}$	Admitted external inputs needed to reproduce derivations through tick T .
$\mathcal{D}_T$	Derivation addresses, rule versions and context required by regenerable states.

Svalbard and the meaning of an anchor

The source first describes locally grounded physical initialization, then names ω as the Svalbard seed-bank baseline [S, pp. 4–5 and 16–18].

Formal convention 2. This edition separates *baseline data* from *coordinate origin*. The name Svalbard identifies the source's baseline role; an actual deployment binds that role to a stated data record. A literal physical-anchor profile additionally supplies its measurements and reference frame. A relational profile uses ω without making its location the origin of every sweep. This is an explicit separation of the source's two anchoring descriptions.

The transition interface

$$F_v : S_T \times e_T \longrightarrow (S_{T+1}, O_T),$$

where e_T is an admitted event and O_T is a finite stream of output words. A realization fixes rounding, word widths, overflow, phase wrapping, lookup versions, branch order and event order in Π_v .

Requirement 1. Every output-affecting input is either part of the state, part of the admitted event, or a fixed manifest value. This supplies the exact meaning of Det-0 reproducibility; the term “0-order” is retained as the source's execution label rather than assigned an unrelated analytic meaning.

3 Log-polar core and OTAN2

Source basis: [S], pp. 4–5, 7–9 and 25–26.

Definition 3. A local log-radius code $k \in \mathbb{Z}$ represents a positive scale through a declared base $b > 1$:

$$\rho(k) = \rho_* b^k, \quad \log_b \left(\frac{\rho(k_2)}{\rho(k_1)} \right) = k_2 - k_1.$$

ρ_* is a local scale unit, not a mandated geographical origin. A radial change is an integer difference in code space. The manifest specifies how k is quantized, bounded and stored or selected by a packed word.

OTAN2 is a phase operator, not an imported arctangent

The source expressly replaces the earlier ATAN2 interpretation with OTAN2 and a two-dimensional radial sweep [S, p. 4]. This specification preserves that distinction.

Formal convention 3: reference phase action. For m phase bins, phase $r \in \mathbb{Z}_m$, local increment $\delta \in \mathbb{Z}_m$, and orientation bit $\eta \in \{0, 1\}$, define

$$\text{OTAN2}_m(r, \delta, \eta) = r + (-1)^\eta \delta \pmod{m}.$$

The increment δ is obtained from a declared LUT relation or admitted phase input. This equation is a minimal concrete realization introduced in this edition. The source does not supply a unique numerical formula for deriving δ from every physical signal.

Self-referential lookup closure

Let κ_v construct a relative lookup key and let I_v interpret a returned word. A step has the form

$$\begin{aligned} q_T &= \kappa_v(X_T, e_T), \\ w_T &= \mathcal{L}_v(q_T), \\ X_{T+1} &= I_v(w_T, X_T, e_T). \end{aligned}$$

The next key is constructed from the updated packed state. Thus the stream feeds its own routing and interpretation, while the lookup and interpretation contracts remain explicit.

Requirement 2. A LUT miss, a phase tie, a radial endpoint and a change of orientation each have a defined outcome. A self-reference also has a progress measure—for example a tick, a finite rewrite depth or an instruction budget—so that each admitted event produces a finite transition.

Resolution shifts

A bit shift changes an integer representation. A physical scale change is the manifest's map from that representation to k , ρ or phase resolution. These maps are recorded separately; the reference phase action does not assume that shifting a log-radius code is the same operation as doubling a physical radius.

4 Klein topology and Lie structure

Source basis: [S], pp. 5, 8–9, 13–14 and 18–20.

Formal convention 4: topological model. One explicit realization of the named Klein-bottle topology is the quotient

$$\mathcal{K} = \mathbb{R}^2 / \sim, \quad (u, v) \sim (u, v + 1), \quad (u, v) \sim (u + 1, -v).$$

Here (u, v) are temporary chart labels for stating the identifications. They do not impose a global physical origin on OTAN2. The second identification supplies the orientation reversal used by the paired representation.

Definition 4. A seam transition transports both the node label and its orientation bit. If a path crosses the reversing seam n times, then

$$\eta' = \eta \oplus (n \bmod 2).$$

Equivalent chart representatives must decode to the same physical or logical state. Topological adjacency and binary-search ordering are specified separately in Section 9.

Continuous transformation layer

The source invokes Lie groups without selecting a particular group or generator. The following phase/reflection model is an explicit choice for this edition:

$$G = (\mathbb{R}/\mathbb{Z}) \rtimes \mathbb{Z}_2, \quad (a, \eta) \star (b, \zeta) = (a + (-1)^\eta b, \eta \oplus \zeta).$$

a and b are phases measured in turns. This expresses local rotation with a possible reversal. It is a local transformation model; it is not a claim that the Klein bottle itself is a Lie group.

Formal convention 5: Det-0 phase algebra. Replacing continuous phase by $\mathbb{Z}_m$ gives the closed finite action

$$G_m = \mathbb{Z}_m \rtimes \mathbb{Z}_2.$$

The continuous design layer and the finite packed realization are thereby related explicitly rather than identified with one another.

Double-packed mirror

For a phase state, define

$$M(r, \eta) = (-r \bmod m, \eta \oplus 1), \quad M(M(r, \eta)) = (r, \eta).$$

A packed mirror also transforms any field whose meaning changes under the seam and recomputes its check bits. RP32 declares those transformations in Section 7.

Requirement 3. Each relevant update U must preserve the declared pair relation. For the reference phase step, this is $M \circ U_\delta = U_\delta \circ M$. Indefinite topological wrapping is represented by repeatable transitions, while the time spent on any one traversal remains governed by a finite execution budget.

5 Fields, axes and cartography

Source basis: [S], pp. 2–3, 6–7 and 16–18.

Lowercase phi and double delta

Definition 5. Given a declared metric d , boundary $\partial\Omega$ and local side convention σ , the boundary field is

$$\phi(x) = \sigma(x) \inf_{y \in \partial\Omega} d(x, y).$$

The boundary, metric, unit and sign convention are part of the field profile. On a reversing seam, the manifest states whether ϕ is a scalar carried unchanged or a locally signed quantity that transforms with orientation.

The source names $\Delta\Delta\phi$ and associates it with shaft θ , structural changes and time T . It supplies no single stencil. This edition therefore keeps the difference direction visible:

$$\Delta_a^2\phi(x) = \phi(x+a) - 2\phi(x) + \phi(x-a),$$

$$\Delta_T^2\phi_T = \phi_{T+1} - 2\phi_T + \phi_{T-1}.$$

A divided-difference profile additionally declares spacing and units. The spatial expression and the temporal expression are different operators; a realization selects the one its output channel denotes.

Psi as the traversal axis

Definition 6. An eigenvector interpretation of Ψ requires a declared operator A_x :

$$A_x\Psi_x = \lambda_x\Psi_x, \quad \Psi_x \neq 0.$$

The source names the eigenvector role but not A_x . The manifest binds that operator, eigenvalue-selection rule, sign/phase convention and degeneracy rule. A packed selector may refer to a precomputed Ψ representative rather than storing a full vector in one channel.

Geometric legenda

Source primitive	Formal role retained in this edition
Apex	A local convergence or phase-accumulation marker; not automatically a global origin.
Pyramid / cone	A selected boundary family, with slope and extent parameters interpreted relative to a local shaft or Ψ relation.
Sphere / circle	An isotropic boundary or footprint family, with declared metric and radius code.
Shaft θ	A local sweep/traversal direction used by a selected difference stencil.
Time T	Logical progression of state and input events, separate from a shape's side-view notation.

Requirement 4. The geometry-to- ϕ map is supplied per primitive. A shape label, salinity value or integrity flag acquires a distance meaning only through that declared map. This keeps the source's cartographic symbolism and numerical boundary evaluation distinguishable.

6 Same-structure packed words

Source basis: [S], pp. 5, 13, 16, 20, 22–23, 25–26 and 31.

Definition 7. The uniform carrier of a w -bit profile is

$$\mathcal{W}_w = \{0, 1, \dots, 2^w - 1\}.$$

A profile supplies a decoder, an interpretation and an encoder. A node word, operator word or footprint word occupies the same carrier; its role is selected by a declared tag and context.

The packet is the operator. In formal terms, a decoded word can select or parameterize its own transition. Same structure means common representation and execution conventions; it does not require every field to have the same meaning for every opcode.

Layouts retained from the source

The conversation presents several distinct layouts. This edition retains them as profiles rather than silently combining their fields.

Source layout	Width	Fields as presented
Early f8 [S, p. 5]	$8 + 12 + 12 = 32$	Log-polar radius; Ψ vector; OTAN2 phase differential.
Uniform spatial [S, pp. 13, 16]	$8 + 8 + 8 + 8 = 32$	OTAN2 phase; Ψ ; ϕ ; XOR/parity or salinity mask.
Twin-agent [S, p. 20]	$32 + 32 = 64$	Two channels, each described as 16-bit OTAN2 plus 16-bit Lie vector.
DIGID [S, p. 22]	$16 + 16 + 32 = 64$	Routing; integrity; payload.
WElip [S, p. 31]	$16 + 16 + 32 = 64$	Ignition/time; wave-phase; payload.

RGBA is a carrier interpretation

The early RGBA mapping uses G for Ψ ; the later source correction gives G a traversal/growth role [S, pp. 23, 25–26]. A stream manifest identifies which interpretation is active. RGBA labels denote packed channels before they denote any display colors or texture coordinates.

Formal convention 6. A profile identifier and version accompany the stream or its enclosing block. Long payloads use finite ordered sequences of uniform words, with length, ordering and continuation rules supplied by the profile. No particular 32-bit field is assigned the meaning of an arbitrarily long lossless payload.

Requirement 5. Identical bytes under different profiles are not treated as the same state. A profile change is an explicit stream event or block boundary, and replay retains the original profile version.

7 RP32 reference encoding

Source basis: [S], 32-bit carrier and RGBA roles from pp. 13, 16 and 25–26.

Formal convention 7. RP32 is the concrete packing convention introduced for this edition. It gives an exact phase/mirror test case while leaving application-specific LUTs and field models as declared inputs.

Bits	Channel	RP32 meaning
0–7	<i>R</i>	Relative phase $r \in \mathbb{Z}_{256}$.
8–15	<i>G</i>	Unsigned traversal/LUT selector $g \in \{0, \dots, 255\}$.
16–23	<i>B</i>	Signed field code $b \in \{-128, \dots, 127\}$, two's complement.
24–31	<i>A</i>	Seven metadata bits and one even-parity bit.

Within *A*, bits 0–2 are the opcode, bit 3 is the jitter flag, bit 4 is orientation η , bit 5 is continuation, bit 6 is a profile-local flag and bit 7 is parity. Opcode values 0–7 name DATA, STEP, GROW, BRANCH, SEAM, EVICT, EMIT, CONTROL, respectively. Their complete application semantics belong to the manifest; the worked trace uses STEP.

Packing, parity and numerical conventions

For metadata $a \in \{0, \dots, 127\}$, define

$$\begin{aligned} v &= r + 2^8g + 2^{16}(b \bmod 256) + 2^{24}a, \\ p &= \text{popcount}(v) \bmod 2, \\ \text{pack}(r, g, b, a) &= v + 2^{31}p. \end{aligned}$$

An intact word has an even total number of set bits. Serialization is least-significant byte first: *R, G, B, A*. The signed field is decoded by subtracting 256 from its byte when that byte is at least 128.

A field-quantization binding may use scale $s > 0$ and

$$Q_s(x) = \min(127, \max(-128, \lfloor x/s + 1/2 \rfloor)).$$

This selects saturation and a stated tie rule. Phase wraps modulo 256; intermediate field differences and products use widened integer arithmetic before a declared rescaling.

Mirror and double packing

The RP32 reference treats *B* as a seam-consistent scalar and *G* as a shared selector:

$$M(\text{pack}(r, g, b, a)) = \text{pack}(-r \bmod 256, g, b, a \oplus 16).$$

Parity is recomputed after the transformation. A double-packed pinion pair is

$$D = W_L + 2^{32}W_R, \quad W_R = M(W_L).$$

A profile using a differently transforming field supplies its own mirror rule rather than reusing this one silently.

8 L-system organogram

Source basis: [S], pp. 10–12, 15–18 and 34.

Definition 8. The generative layer is a deterministic, parameterized rewriting system

$$\mathcal{G}_v = (\Sigma, w_0, \mathcal{P}_v, J_v), \quad w_0 = \text{EncodeAxiom}_v(\omega).$$

Σ is the alphabet, $\mathcal{P}_v$ the production rules, and J_v the interpreter mapping a finite derived word into packed operators. This distinguishes the baseline data ω from the initial grammar word w_0 , both of which are called an axiom in different source passages.

Source symbol	Interpretation contract
F	Advance along the current Ψ /phase relation and emit or extend a segment.
$+, -$	Apply a positive or negative OTAN2 increment selected by the active LUT.
$[,]$	Push and restore a complete local branch context, including orientation.
Apex / sphere	Select a terminal or footprint boundary primitive and its field profile.
$\ll, \gg$	Invoke a declared resolution transition, with explicit integer-width semantics.
XOR	A declared bitwise operation or condition; parity evaluation and structural mutation are distinct uses.

Time- and input-conditioned rewriting

For a symbol a and its permitted context c , a rule produces a unique finite string:

$$\mathcal{P}_v(a; c, \omega, T, \mathcal{E}_{\leq T}) \in \Sigma^*.$$

Formal convention 8. The reference grammar contract uses parallel rewriting of one finite generation at a time. A realization records rule priority, branch ordering, maximum expansion depth, stack bounds and the interpretation of an unmatched symbol. This convention supplies an execution order not fixed in the source.

Derivation identity and regeneration

A generated node is addressed by its baseline version, grammar version, original tick and finite derivation path σ :

$$N_{T,\sigma} = J_v(\text{Derive}(w_0, \mathcal{P}_v, T, \mathcal{E}_{\leq T}, \sigma)).$$

The derivation path records which production and branch generated the node. It need not be a global physical coordinate.

Requirement 6. Regenerable nodes have a well-founded derivation and retain access to every non-derived input on which that derivation depends. Exact replay uses the node's original tick and rule version, not the current request time. The source gives operator roles but no complete salinity-to-branch production table; that table is a realization binding.

9 f8 indexing and Hadamard routing

Source basis: [S], pp. 4–5, 11–14 and 17–18.

Middle-out binary search

Definition 9. f8 is the paradigm’s relational indexing layer over generated packed states. A realization supplies a total ordering key such as

$$K_v(n) = (k_\Psi(n), k_\rho(n), k_\theta(n), k_{\text{derivation}}(n)),$$

compared lexicographically within a stated epoch. The components are local traversal and derivation labels; they do not require a global geographic grid.

Formal convention 9. “Middle-out” is realized by selecting a median key from an ordered finite active set, then recursively doing the same in each half. This is a concrete interpretation of the source’s symmetric expansion language, not a production rule supplied by the source itself.

Requirement 7. A canonicalization rule makes equivalent topological representatives index to the same logical object. Distinct logical objects receive a deterministic tie-break key. A change in the Ψ ordering either starts a new index version or performs a recorded rebuild.

Topological neighbors are defined by $\mathcal{N}_v$; search-tree children are defined by the key ordering. A tree edge is therefore not automatically a physical adjacency or a Klein-bottle seam. Maintaining these two relations explicitly gives precise meaning to the source’s “Klein Bottle BST.”

Hadamard operator

The source specifies component-wise multiplication for routing [S, pp. 13–14]. Here it is applied to decoded numerical lanes, not to an undifferentiated packed integer.

Definition 10. For lanes u_i, v_i in a fixed-point scale 2^f , define

$$H_f(u, v)_i = \text{clip}\left(\text{round}_v\left(\frac{u_i v_i}{2^f}\right)\right).$$

The manifest fixes intermediate width, rounding and clipping. A boundary-driven candidate is

$$q = H_f(\text{DecodeVector}(W), \nabla_v \phi), \quad n' = \text{SelectNeighbor}_v(n, q, \mathcal{N}_v).$$

$\nabla_v \phi$ is a declared discrete field evaluator. The selection rule supplies admissibility, ties, stopping and fallback behavior.

Requirement 8. Parity, opcode and continuation bits are handled by their own operations. They are not multiplied as if they were vector components. A routing shortcut is accepted only through the profile’s stated boundary and adjacency tests; path quality and traversal cost are then measured on that defined routing algorithm.

10 FIFO eviction and regeneration

Source basis: [S], pp. 13–18, 22, 26 and 34–37.

Definition 11. Let $Q = [q_0, \dots, q_{n-1}]$ be an oldest-first FIFO of capacity $C \geq 1$. Insertion is

$$\text{push}_C(Q, x) = \begin{cases} [q_0, \dots, q_{n-1}, x], & n < C, \\ [q_1, \dots, q_{C-1}, x], & n = C. \end{cases}$$

The second case emits an eviction event naming q_0 . The implementation can use a ring position; physical array movement is not part of this definition.

What eviction removes

Formal convention 10. The active queue stores derived words or double-packed pairs. Eviction removes their active copies. Baselines, production rules, derivation descriptors and admitted input records have declared retention policies separate from the queue.

For w -bit queue items,

$$M_{\text{active}} = C w / 8 \text{ bytes},$$

while a complete realization also accounts for

$$M_{\text{total}} = M_{\text{active}} + M_{\text{baseline}} + M_{\text{rules}} + M_{\text{inputs}} + M_{\text{index}} + M_{\text{runtime}}.$$

This defines exactly which memory is bounded by the FIFO and makes the source's eviction objective measurable.

Regeneration contract

A regeneration request names a derivation descriptor

$$d = (v, \omega_{\text{id}}, T, \sigma, \mathcal{E}_{\text{required}}),$$

and evaluates

$$\text{Regen}(d) = \text{Eval}_v(\omega, \mathcal{P}_v, T, \sigma, \mathcal{E}_{\text{required}}).$$

The result is checked against the same profile, field conventions and mirror relation as an originally emitted word.

Requirement 9. An external measurement or arbitrary packet that cannot be derived from the retained baseline and rules is retained, supplied again, or explicitly excluded from the exact-regeneration set. Cache eviction and removal of the only copy of an input are different transitions.

Capacity changes and paired nodes

A reduction from C to C' retains the newest $\min(|Q|, C')$ items and evicts older ones in order. If the item is a double-packed pinion pair, both channels are one eviction unit. Active references use derivation identities rather than raw queue positions, so a recycled slot is not mistaken for its previous occupant.

Requirement 10. The manifest defines regeneration budget, cache-miss handling and the behavior of a request whose required derivation inputs are unavailable. Those outcomes are explicit states, not silently fabricated reconstructed data.

11 Mirrored agents and time

Source basis: [S], pp. 18–20, 24, 26 and 31.

Definition 12. A telepresent pair consists of agents A and B with a declared mirror map M and relation

$$X_T^B = M(X_T^A).$$

The source’s “double-nut-sack mirror symmetry” is retained as the name of this dual-channel motif. The formal term used below is *mirror pair*. The double-packed pinion is its joint representation.

Equivariant updates

For a shared phase increment δ , the RP32 update is

$$U_\delta(r, \eta) = (r + (-1)^\eta \delta \bmod 256, \eta).$$

Because

$$M \circ U_\delta = U_\delta \circ M,$$

independent application of the same increment maintains the pair relation. General agent operations either obey this equation or declare the conjugate right-channel update

$$U^B = M \circ U^A \circ M^{-1}.$$

A right bit shift is not assigned the inverse meaning of a left bit shift unless the profile preserves the bits needed for that inverse. The mirror is the stated transformation, not an assumed consequence of shift direction.

Pair comparison and parity

A full-word mirror comparison is

$$D_T = X_T^B \oplus M(X_T^A), \quad D_T = 0$$

for equal-width words. This checks the declared relationship of all compared bits. Even parity checks a different invariant: the number of set bits modulo two. The two checks have separate status fields in an agent’s output.

Logical synchronization

Formal convention 11. Agent synchronization means equality of the version, baseline, admitted input prefix, tick and transformed state. A distributed realization declares how new inputs are delivered and ordered. Identical initial states and identical ordered inputs allow independent deterministic replay; the time required to deliver a new input is not removed from the model.

Requirement 11. A synchronization decision names an epoch and input-prefix identifier. Clock wrap, a delayed event, a conflicting event order and a rule-version change each have a defined resolution. The source’s unilateral or globally synchronized descriptions become conditional execution requirements under this explicit input model.

Juicyness / juiciness

The source’s game-development feedback term denotes an output mapping from phase, jitter and field changes to animation, deformation or haptic parameters. That mapping is versioned independently of the canonical packed state unless its effects are themselves admitted as new inputs.

12 LUS, DIGID and UU ID

Source basis: [S], pp. 1, 9 and 20–22.

LUS emission

The source uses LUS as “Linear Unit Strain” and later extends the same stream to general telemetry. A profile declares whether its output denotes physical strain, simulated boundaries, agent state or a digital footprint.

Definition 13. A LUS record envelope is

$$\ell = (v, \omega_{\text{id}}, T, s, \text{kind}, W^*, \text{status}),$$

where s is a sequence number and W^* is a finite word sequence. The envelope’s representation is supplied by the transport profile; it is not implicitly squeezed into the four RP32 bytes.

Identity as a derivation relation

DIGID and UU ID are the source’s names for extending a spatial node to any packet, log or digital footprint [S, pp. 20–22].

Formal convention 12. A reference identity is a structured name

$$\text{ID}(n) = (v, \omega_{\text{id}}, \text{producer}, \text{epoch}, s, \sigma).$$

The namespace rules define when two such names refer to the same derivation. The identity is linked to the packed phase trajectory, while producer and sequence allocation supply explicit disambiguation for external packets.

Relation	Required meaning
Naming consistency	The same identity resolves to the same baseline, input context and derivation.
Representation integrity	The selected parity and full mirror checks hold for the received representation.
Origin / admission	A deployment-specified predicate determines which producer and input are admitted.
History reproducibility	Retained derivation inputs regenerate the same record under its original version.

The source provides no complete issuer-authentication or origin-verification algorithm. Those predicates remain explicit bindings in the manifest rather than being equated with parity, topology or physical location. The source’s rejection of SHA256 *as a spatial seed* is preserved; this edition introduces no hash seed into ω or OTAN2.

Arbitrary payloads

For payload bytes P and profile payload capacity q bytes per fragment, the profile emits an ordered fragment sequence of length $\lceil |P|/q \rceil$ together with its length and ordering metadata. A generated payload may instead be represented by a replay descriptor when its inputs and decoding rule are retained.

Requirement 12. Equality of identifiers, equality of bytes, origin admission and successful regeneration are separately defined results. A LUS or DIGID consumer can therefore determine which property a record actually reports.

13 WElip forward-time profile

Source basis: [S], pp. 30–32 and 34–37.

The source names WElip / Reverse Electronic Warfare (REW) as a forward-only, execution-ledger extension. It also uses “PRISM reverse lock,” “turn-key ignition” and “poisoned pill.” These labels are retained below as protocol roles.

Formal convention 13: interaction profiles. The source contains both on-demand historical regeneration and later query-free operation. This edition keeps two explicit profiles:

R: regenerative	Admits a named regeneration request for a retained derivation at its original tick.
W: WElip	Admits forward ignition, progression and local lifecycle events. It exposes no historical-read operation in its public event alphabet.

A realization chooses a profile or states a controlled boundary between them. Historical replay is not simultaneously treated as an ordinary allowed call and an automatic violation of the same interface.

Forward event semantics

For WElip, let

$$\mathcal{V}_W = \{\text{IGNITE, ADVANCE, RESIZE, INVALIDATE, EMIT}\}.$$

Ignition admits a payload and binds its baseline, epoch and phase context. **Advance** applies the next deterministic operator. **PRISM reverse lock** names the event-admission rule that rejects a backward-history operation at this interface. **Emit** produces the next record in sequence.

Definition 14. Forward-time order requires $T_{\text{next}} > T_{\text{current}}$ for a new logical transition. Multiple records at one tick use a strictly increasing within-tick sequence. A finite time field is paired with an epoch so wrap does not reverse logical order.

Poisoned-pill eviction as a state transition

Formal convention 14. The source’s poisoned-pill operation is formalized as *localized invalidation and FIFO eviction*. The manifest selects the affected active slot set, records the cause, invalidates its active entries and permits subsequent forward regeneration only under the selected profile.

$$\text{Invalidate}_J(Q) = Q \text{ with the declared active entries } J \text{ removed.}$$

This defines the local memory action; the source supplies no additional machine-level mechanism for its broader labels.

Requirement 13. Detection of an invalid event has an explicit interface predicate. A read attempt is an event only when the interface detects and reports it; a change in parity is not presumed merely because observation occurred. The retained baseline and unaffected queue partitions have their own declared transition rules.

14 RGBA realization and scaling

Source basis: [S], pp. 23–26 and 35–37.

Semantic channels before physical storage

The source’s correction on page 25 is decisive: the log-encoded polar LUT remains the architecture. RGBA is an encoding of its packed state, not a replacement by a Cartesian image model.

Definition 15. A realization adapter maps a semantic word stream to a storage or execution medium:

$$\text{EncodeAdapter}_\alpha : \mathcal{W}^* \rightarrow \mathcal{B}_\alpha, \quad \text{DecodeAdapter}_\alpha \circ \text{EncodeAdapter}_\alpha = \text{id}$$

on the supported word domain. $\mathcal{B}_\alpha$ may be a register stream, a byte buffer or a GPU-resident channel representation.

Requirement 14. An exact adapter preserves channel bit patterns, order, word boundaries and version. A display transformation, interpolation or reduced-precision conversion is an explicitly different output view, not the canonical execution state.

Compute availability as an admitted control input

Formal convention 15. Let B_T be a declared compute budget and let

$$\pi_v(B_T) = (C_T, d_T, m_T, \tau_T)$$

select FIFO capacity, derivation depth, phase resolution and scheduling quantum. Capacity changes follow Section 10; changes to depth or phase interpretation are versioned state transitions.

If compute availability changes semantic outputs, its control events are part of the replay input. If it changes only cache residency, the same requested derivation must regenerate the same state regardless of cache size.

Chained LUT textures

A cascade is a finite composition

$$\mathcal{L}_v^{[k]} = \mathcal{L}_{v,k} \circ \dots \circ \mathcal{L}_{v,1},$$

with declared intermediate word profiles. Increasing available compute may extend the active cascade or refine a generation budget without changing the foundational word-and-lookup model.

Requirement 15. Each cascade stage states its input and output profile, execution cost budget and progress rule. A deployment specifies partitioning, input delivery, admission order and resource ownership before interpreting matching local states as distributed synchronization.

Deployment roles retained from the source

The source names datacenter compute, GPU pipelines, FPGAs, microcontrollers and Digital Realty as intended environments [S, pp. 23–24, 36]. Here they remain deployment targets. Their concrete realization is expressed through adapter conformance and measured throughput, latency, memory and replay behavior, rather than assumed from the infrastructure name.

15 Wave and sensory adapters

Source basis: [S], pp. 27–29 and 33.

The source extends the same LUT fabric to cicada cascades, micro-saccades, IPD, IDT, FMCW and LRAD. Their formal place is an input/output adapter layer around the canonical packed transitions.

Source term	Declared interface quantity
IPD	Relative phase between paired observations, with phase unit, wrapping and ambiguity rule.
IDT	Difference in arrival or sample times, with clock and time-unit convention.
FMCW	Input record of sampled chirp/return parameters and a selected interpretation model.
LRAD	Named acoustic-medium integration role; a deployment supplies its transduction interface.
Micro-saccades	Small deterministic sampling or phase perturbations linked to the jitter channel.
Cicada cascades	Periodic scheduling family used to trigger sampling, regeneration or eviction work.

Paired observational features

Formal convention 16. For measured phase codes p_L, p_R and timestamps t_L, t_R , define adapter features

$$\Delta p = \text{wrap}_m(p_R - p_L), \quad \Delta t = t_R - t_L.$$

Their map to OTAN2 increments, field values or growth selectors is a versioned LUT binding. A conversion to physical direction, range, density or strain additionally declares geometry, timing calibration and the signal model used by that adapter.

Cicada scheduling and one-bit jitter

For declared periods $p_1, \dots, p_k$, a simple finite scheduler is

$$c_i(T) = \mathbf{1}[T \bmod p_i = 0].$$

The source's 13/17-cycle example can be represented by $p_1 = 13, p_2 = 17$. Both triggers coincide at multiples of 221; the manifest therefore states what happens when several scheduled events are due together.

A reference one-bit perturbation is

$$j_T = (c_1(T) + c_2(T)) \bmod 2.$$

This is an explicit choice for the edition, not a formula supplied by the source. Its effect on phase or sampling is declared separately from the integrity check. Parity is recomputed after an intentional jitter update.

Requirement 16. Observed samples, calibration versions and scheduling events that affect canonical state are retained in the replay context. Wave mixing, Hadamard lane multiplication and byte-integrity checking remain separately typed operations. The source gives no numerical transfer function that makes them interchangeable.

16 Transition contract

Source basis: [S], execution-stack descriptions on pp. 12, 14, 33–34 and 36–37.

Definition 16. One admitted transition is a finite evaluation of the following ordered contract. All named functions are versioned by the manifest.

Step	Operation	Contract
1	Admit	Check interaction profile, epoch, sequence and input kind.
2	Decode	Read the word profile and produce typed phase, traversal, field and metadata values.
3	Resolve context	Bind baseline, rule version, local topology and the required derivation inputs.
4	Lookup / grow	Select an OTAN2 increment or a finite L-system expansion through the self-referential LUT.
5	Evaluate	Apply field, Hadamard, phase and branch operators to their typed lanes.
6	Transport	Apply seam and mirror maps; canonicalize topological labels.
7	Repack / check	Encode with stated rounding and widths; recompute parity; compare the mirror relation.
8	Retain / evict	Insert an accepted active item, or apply the selected local invalidation event.
9	Emit / progress	Output status and LUS/DIGID words; advance logical tick or within-tick sequence.

Result classes

A transition returns a typed result rather than conflating every failure with one bit pattern:

$$\text{Result} \in \{\text{ACCEPT}, \text{DEFER}, \text{MISS}, \text{INVALID}, \text{EVICTED}\}.$$

Accept means the named checks passed. **Defer** means the execution budget postpones work without changing its derivation identity. **Miss** means a needed lookup or replay dependency has no declared value. **Invalid** names an admission or representation violation. **Evicted** records a lifecycle operation.

Reference execution skeleton

```
e = admit(profile, next_event)
x = decode(e.words, e.version)
k = relative_key(x, baseline, local_context)
o = lookup_or_derive(k, rules, retained_inputs)
y = execute_typed(o, x)
y = transport_and_mirror(y, topology)
w = repack_and_check(y)
Q, events = apply_lifecycle(Q, w, capacity, profile)
emit(version, logical_time, w, events)
```

The skeleton is a contract decomposition. Appendix B gives executable arithmetic for RP32 phase stepping, mirror construction and FIFO replay; the full application binds the other functions through Appendix D.

17 Derived properties

Formal deductions from this edition's definitions.

P1. Conditional deterministic replay

Statement. Two runs with identical initial state, manifest, baseline and ordered event sequence produce identical outputs, provided each transition function is single-valued.

Derivation. The initial states agree. If the states agree at tick T , the same deterministic transition applied to the same event gives the same state and output at tick $T + 1$. Induction supplies equality for every finite tick. Word width alone is not a premise sufficient to establish this result; the complete transition inputs are.

P2. Bounded active FIFO

Statement. For fixed $C \geq 1$, repeated application of push_C preserves $|Q| \leq C$.

Derivation. An insertion below capacity increases the length by one. An insertion at capacity removes one item and appends one. The corresponding active storage is exactly $Cw/8$ bytes for a preallocated w -bit item array, apart from its fixed management state.

P3. Exact regeneration of derived states

Statement. Evicting an active copy does not alter the result of regeneration when the original derivation descriptor and every required input remain available.

Derivation. Regeneration evaluates the same versioned expression with the same arguments. By single-valued evaluation, the result equals the original. This property concerns the defined regeneration set, which includes its input-retention condition.

P4. Reference mirror preservation

Statement. The reference phase step preserves $X^B = M(X^A)$.

Derivation. For $s = (-1)^\eta$,

$$M(U_\delta(r, \eta)) = (-r - s\delta, \eta \oplus 1),$$

while

$$U_\delta(M(r, \eta)) = (-r + (-s)\delta, \eta \oplus 1).$$

The results agree modulo m . The RP32 packing map is deterministic and recomputes parity, so the packed relation also agrees. Applying M twice restores the phase and orientation.

P5. Representation independence

Statement. If an adapter's encode/decode composition is the identity and its operations preserve the same transition semantics, changing the storage medium does not change canonical output words.

Derivation. Decode recovers the same semantic input. The same transition produces the same output, which can be encoded again without changing its decoded value. This gives a precise criterion for realizing the log-polar architecture through different physical carriers.

18 Conformance and checks

Conformance criteria for the architecture and definitions in Sections 1–17.

Conformance matrix

Test	Object	Pass criterion
T1	Encode/decode	Every tested supported field tuple is recovered exactly.
T2	Phase wrap	$r + \delta$ and orientation-reversed steps agree with declared modular arithmetic.
T3	Mirror	$M^2(W) = W$; reference updates commute with M .
T4	Integrity	All single-bit word changes fail RP32 even parity; full mirror equality is tested separately.
T5	Topological seam	Equivalent representatives yield identical logical identities and transported field meaning.
T6	FIFO	Capacity is never exceeded; eviction order and pair atomicity match the manifest.
T7	Replay	Original and regenerated words agree byte-for-byte for retained derivations.
T8	Word profiles	Version mismatch is explicit; profiles are not reinterpreted silently.
T9	Scaling / agents	Equal semantic input histories yield equal canonical outputs under tested resource schedules.
T10	Application fields	Calibrated test inputs produce the profile’s declared physical or simulated outputs.

RP32 arithmetic checks executed for this edition

The Appendix B arithmetic was exercised with phase $r = 0, \dots, 255$, both orientation states, selector $G = 3$, field $B = -7$ and the STEP opcode. The recorded checks are:

Check	Passing cases
Mirror involution over all phase/orientation states	512
Step/mirror commutation for all 256 increments	131,072
Single-bit corruption checks on the three left words in Appendix A	96
Three-state replay and capacity-two FIFO trace	1

These are checks of the stated RP32 arithmetic and trace. The source-specific physical baselines, general growth rules, full f8 implementation, network adapter and wave interpretation are separate conformance objects identified by their own manifests and tests.

Performance record

A benchmark records word profile, LUT size, rule set, queue capacity, retained-input size, hardware adapter, event distribution, throughput, latency, regeneration cost and output agreement. A comparison with another spatial index uses the same workload and accuracy target. This turns the source’s speed, memory, scaling and replacement objectives into reproducible engineering measurements.

Appendix A Worked packed-state trace

Carrier and mirror motifs: [S], pp. 13, 16 and 19–20. Numerical example: this edition.

The reference state starts with phase $r = 250$, selector $G = 3$, signed field $B = -7$, STEP opcode 1, orientation 0 and all other metadata flags zero. The right word is always the declared mirror of the left word. Two shared increments, 11 and 9, produce the following trace.

Tick	Input	Left R	Right R	Meaning
0	Initial	250	6	$-250 \equiv 6 \pmod{256}$
1	$\delta = 11$	5	251	Forward phase wraps; mirrored phase moves oppositely.
2	$\delta = 9$	14	242	Both channels preserve the mirror relation.

Exact 32-bit words

Tick	Left word, hexadecimal	Right word, hexadecimal
0	81F903FA	11F90306
1	81F90305	91F903FB
2	01F9030E	91F903F2

The highest bit of each word is recomputed even parity. Consequently the top hexadecimal byte can change even when the seven metadata bits do not.

Double-packed 64-bit pinion words

The displayed high half is the right word; the low half is the left word:

$$\begin{aligned} D_0 &= 0x11F9030681F903FA, \\ D_1 &= 0x91F903FB81F90305, \\ D_2 &= 0x91F903F201F9030E. \end{aligned}$$

The first pair serializes, least-significant byte first, as

$$FA\ 03\ F9\ 81\ 06\ 03\ F9\ 11.$$

Capacity-two eviction and replay

Starting empty, the queue evolves as

$$[] \rightarrow [D_0] \rightarrow [D_0, D_1] \rightarrow [D_1, D_2].$$

The final insertion evicts D_0 as one complete pair. Retaining the initial tuple, profile version and event list [11, 9] allows recomputation of every displayed word. Regenerating tick 0 evaluates the original initial tuple; it does not use tick 2 as a substitute for the original context.

What this trace establishes. The declared phase wrapping, mirror transformation, parity reconstruction, pair packing and FIFO policy fit together exactly in the RP32 reference case.

Appendix B Reference arithmetic

Implementation of this edition's RP32 definitions.

```
from collections import deque

def pack(r: int, g: int, b: int, a: int) -> int:
    if not (0 <= r < 256 and 0 <= g < 256
            and -128 <= b < 128 and 0 <= a < 128):
        raise ValueError("RP32 field out of range")
    v = r | (g << 8) | ((b & 255) << 16) | (a << 24)
    return v | ((v.bit_count() & 1) << 31)

def unpack(w: int) -> tuple[int, int, int, int]:
    if not (0 <= w < 2**32) or w.bit_count() & 1:
        raise ValueError("Invalid RP32 word or parity")
    r, g, b = w & 255, (w >> 8) & 255, (w >> 16) & 255
    return r, g, b - 256 if b >= 128 else b, (w >> 24) & 127

def mirror(w: int) -> int:
    r, g, b, a = unpack(w)
    return pack((-r) % 256, g, b, a ^ 16)

def step(w: int, delta: int) -> int:
    if not (0 <= delta < 256):
        raise ValueError("Phase increment out of range")
    r, g, b, a = unpack(w)
    sign = -1 if (a & 16) else 1
    return pack((r + sign * delta) % 256, g, b, a)

def pair(w: int) -> int:
    return w | (mirror(w) << 32)

def replay(deltas: list[int]) -> list[int]:
    w = pack(250, 3, -7, 1)
    out = [pair(w)]
    for delta in deltas:
        w = step(w, delta)
        out.append(pair(w))
    return out

trace = replay([11, 9])
assert trace == [0x11F9030681F903FA,
                 0x91F903FB81F90305,
                 0x91F903F201F9030E]

fifo = deque(maxlen=2)
for item in trace:
    fifo.append(item)
assert list(fifo) == trace[-2:]
assert replay([11, 9]) == trace
```

RP32 words are the function inputs and outputs. The 64-bit pair is a separate carrier, with its left and right halves retaining their own parity. The replay seed above is the explicitly declared initial tuple, not a cryptographic hash or an asserted physical Svalbard measurement.

Appendix C Source concordance

[S] Source document. *fault lines seismic game input output LUS log encoded polar LUT kinematic calculus mosTAD-POLE (the green one)*. Supplied PDF, 38 pages. Page references below use the source’s own exported page numbering.

Source pp.	Source contribution retained	This edition
1–3	mosTADPOLE input/output; fault-line, salinity and strain examples; SDF, integer and parity vocabulary.	§§2, 5, 7, 12
4–5	Author’s OTAN2 correction; no fixed physical origin; log-polar LUT; Ψ ; f8; physical distribution rather than SHA256 seed.	§§1–4, 9
6–7	Pyramid/cone, sphere/circle, apex and cartographic legenda.	§5
8–9	Topological Fixed-Point Virtual Machine (TPVM); Det-0 and LUS stack.	§§1–3, 12, 16
10–12	L-system organogram, grammar alphabet and growth interpretation.	§8
13–16	Uniform packed operators; Hadamard routing; FIFO drop table; eviction and regeneration.	§§6–10
16–18	ω as Svalbard baseline; shaft θ ; $\Delta\Delta\phi$; middle-out BST; double-packed pinion; T .	§§2, 5, 8–10
18–20	Mirror symmetry; telepresent agents; Lie-group tracking; juiciness; twin 32-bit channels.	§§4, 7, 11
20–22	Any packet or log as DIGID / UU ID; 16/16/32-bit footprint layout.	§§6, 12
23–24	Datacenter deployment and initial RGBA mapping.	§§6, 14
25–26	Author’s centrality correction: the architecture is the self-referential log-encoded polar LUT; later RGBA roles and chained scaling.	§§1, 3, 6, 14
27–29	Cicada cascades; micro-saccades; IPD/IDT; FMCW/LRAD integration vocabulary.	§15
30–32	WElip / REW; PRISM reverse lock; ignition; forward-only interaction; poisoned-pill eviction.	§13
33–34	Consolidated universal stack and operator inventory.	§§1, 16–18
35–38	Deployment, interaction and repeated execution loop.	§§12–18; Apps. A–D

Terminology preserved across stages

TPVM, Generative Topological Fabric, Packed Topological FIFO Matrix, Uniform Packed Operator Paradigm, DIGID, WElip and the final spatial-ledger descriptions are treated as successive source names or functional views. This edition’s title centers the author’s explicit page-25 correction rather than replacing those terms with a new unrelated architecture.

Formal additions made visible

The quotient model, finite phase action, RP32 metadata allocation, explicit mirror map, parameterized grammar contract, median-based middle-out interpretation, event profiles and worked numerical trace are formalization choices introduced in this edition. Source descriptions and these added definitions remain distinguishable throughout the document.

Appendix D Realization manifest

A realization completes the parameter bindings below. The supplied conversation provides the named architecture and qualitative operator roles; it does not provide the numerical or algorithmic contents of every binding.

Binding	Required record
Baseline ω	Identifier, version, contents, units and any literal physical-anchor interpretation.
OTAN2 / log-polar LUT	Phase modulus, log base, scale unit, increment derivation, lookup entries, missing-key behavior.
$\phi, \Delta\Delta\phi, \Psi$	Metric, boundaries, units, difference stencil, eigen-operator and selection/tie rules.
Grammar and f8	Alphabet, production table, rule priority, depth bounds, derivation-address format and ordering key.
Packing and mirror	Profile/version, widths, rounding, overflow, byte order, continuation rules and complete seam map.
Hadamard routing	Numerical lanes, fixed-point scale, intermediate width, neighbor selection and stopping rule.
Eviction / replay	Active capacity, atomic item, retained-input policy, regeneration budget and dependency-miss result.
Time and agents	Epoch, input order, producer namespace, delivery model, pair relation and reconciliation rule.
Identity / WElip	Origin-admission predicate, R or W interaction profile, detection events and invalidation scope.
Adapters and tests	Carrier round-trip rules, physical calibration when used, conformance vectors and benchmark record.

Edition definition

This document formalizes the paradigm as a versioned, relational state machine whose center is the self-referential log-encoded polar LUT. It preserves the author’s operator vocabulary and architecture stages; supplies explicit mathematical contracts; and provides a concrete packed phase/mirror realization with an exact eviction-and-replay trace.

PARADIGM ATTRIBUTION / IDENTIFIER STAMP

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The Self-Referential Log-Encoded Polar LUT Paradigm

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